

Temperature Kills Models, Not Answers: Decoding Robustness Is a Model Property

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Abstract

Truncation samplers such as top- p , min- p , and top- $n\sigma$ exist to prevent the accuracy collapse of high-temperature sampling. We ask whether the collapse itself is universal. Using a single controlled pipeline, we evaluate 7 instruction-tuned models from 5 vendors (1.7B to 12B parameters), 4 quantization levels, 8 sampler arms, and 2 tasks, totaling roughly 200,000 generations. At deployment temperatures ($T \leq 1.3$), sampler choice barely matters on six of the seven models: the per-model accuracy spread across samplers is 1.3 to 10.0 percentage points (pp), an upper bound by construction. The exception is Llama-3.1-8B, which loses 34 pp (GSM8K) and 41 pp (MMLU-Pro) under plain temperature sampling at $T=1.3$, while every other model, including the older Mistral-7B-v0.3 and the accuracy-matched Qwen3-1.7B, stays within 8 pp. The collapse is a failure to terminate, not a failure to reason: accuracy conditional on producing a well-formed answer stays at the greedy level in every cell (97% for Llama on GSM8K at $T=1.3$). A temperature ladder up to $T=2.0$ locates each model’s cliff and shows that truncation samplers earn their keep only past it. A logit probe and a forced-token perturbation experiment trace the fragility to a self-reinforcing instability: one off-distribution token in context multiplies the fragile model’s chance of drawing the next one. Practitioners tuning a current model can leave the sampler alone.

1 Introduction

The truncation-sampler literature, from top- k and nucleus sampling to min- p and top- $n\sigma$ [1, 6, 8, 12], is motivated by a single failure mode: raise the sampling temperature and output quality collapses. A recent critique [10] questioned whether the reported gains of newer samplers survive proper statistical controls. We ask the prior question: **is the collapse itself a property of decoding, or of the model?**

Our contributions:

- **A single-pipeline factorial study.** Seven instruction-tuned models, four GGUF quantization levels plus a BF16 anchor, six decoding methods plus two chain-order ablation arms, $T \in \{0.7, 1.0, 1.3\}$, on GSM8K and MMLU-Pro with $N=50$ items and $K=3$ repetitions, about 202,000 generations on commodity GPUs (RTX 2080 Ti cluster nodes and one RTX 5070 Ti workstation), with item-clustered bootstrap confidence intervals throughout.
- **Robustness is model-bound.** Llama-3.1-8B is the lone collapse (-34/-41 pp at $T=1.3$). Mistral-7B, Qwen2.5-7B, Qwen3-1.7B/4B/8B, and Gemma-3-12B all hold within 8 pp. On the robust models the sampler is worth at most a few points, and that estimate is an upper bound because the spread statistic is biased upward.
- **The collapse is degeneration, not wrong answers.** Conditional on emitting a well-formed final answer, accuracy is flat in temperature everywhere. What varies is the probability of derailing into unparseable text mid-generation. The standard GSM8K flexible parser

disguises this failure mode as wrong answers; the MMLU-Pro letter parser reports it as parse failure.

- **A cliff map from $T=1.3$ to 2.0.** Every model has a cliff; its location is the model property, a staircase one rung apart (on GSM8K: Llama collapses by $T=1.5$, Qwen2.5 by 1.7, Gemma-3 by 2.0; all three pairwise disjoint CIs by 1.7). Truncation samplers shift the cliff right in order of tail-cutting strength ($\text{top-}n\sigma \gg \text{min-}p \gg \text{top-}p$). This reconciles our null with the sampler papers: their gains are real but live past the cliff, where no one deploys.
- **Supporting results.** Quantization (Q8 to Q3) neither causes a collapse nor reorders samplers; its effect grows as models shrink (-9 pp on Qwen3-1.7B, null at 7B and above). Self-consistency shows robust models sample diverse token sequences that converge to the same answer. A logit probe on each model’s greedy path finds per-token uncertainty ranks with fragility (Spearman $\rho = 0.96$); generation-length censoring shows derailment is absorbing (no model derails and recovers); Llama and Mistral tie on every distributional statistic, including the temperature-scaled beyond-top-20 hazard; and a forced-token experiment isolates the difference: one injected tail token multiplies Llama’s subsequent escape hazard 2.3-fold while every robust model absorbs it ($\rho = +0.86$ with the drop).

2 Related work

Truncation samplers. Top- k [1] and nucleus sampling [6] were introduced to prevent *neural text degeneration*: incoherent continuations caused by sampling from the unreliable tail of the distribution. Successors refine the truncation rule: typical sampling [7], eta and epsilon sampling framed as desmoothing [5], the confidence-scaled threshold of min- p [8], and the logit-space cut of top- $n\sigma$, temperature-invariant by construction [12]. Welleck et al. [13] formalized the limiting case of degeneration: standard decoders can fail to terminate at all, a failure mode of the decoding process rather than of answer selection. Our mechanism result closes a loop with this lineage: the high-temperature *accuracy* collapse on reasoning benchmarks is exactly this failure (generations that run to the token cap, not wrong answers), and on most current models it has moved past deployment temperatures.

Reproducibility of sampler gains. The min- p critique [10] reanalyzed nine models across 31 temperatures and found min- p “largely indistinguishable” from other samplers once hyperparameter volume is equalized, tracing part of the reported gains to a change in temperature ordering. We adopt its methodological recommendations (statistical uncertainty, explicit temperature-first and temperature-last arms) and answer the question it left open: it pooled across models to ask whether min- p is special (it is not); we ask whether the collapse the samplers exist to fix is itself model-dependent (it is). Garces Arias et al. [3] is the closest systematic hyperparameter sweep, on open-ended generation rather than verifiable tasks. Song et al. [11] documents per-model gaps between greedy decoding and sampling but frames them as evaluation noise to report, not as a model property to explain.

Robustness to decoding randomness. Renze and Guven [9] sampled temperatures from 0.0 to 1.6 on multiple-choice problem solving and found no statistically significant effect between 0.0 and 1.0, across models, prompting techniques, and domains. Our deployment-band null replicates this, and our ladder explains why the null is incomplete: the per-model heterogeneity starts just past where pooled analyses stop looking, and pooling hides the one model that falls early. The

nearest neighbor is Grover [4], which concludes that instruction tuning is the primary determinant of robustness to decoding randomness. Our data refine this claim: all seven of our models are instruction-tuned and one still collapses, so instruction tuning is necessary but not sufficient. Their design also cannot observe our phenomenon: they cap the sweep at $T=1.0$ while the Llama collapse happens between 1.0 and 1.3; their robust models score 6 to 30% on synthetic tasks, so there is little room to fall; their instruct-versus-base contrast is confounded with 4-bit versus full-precision inference, whereas we sweep a clean quantization axis and find quantization is not the variable; and they test no truncation samplers. Fastowski et al. [2] connect token-distribution entropy to temperature sensitivity on closed-book QA and report that larger models are more stable. Our data support the entropy link but not the size story: Qwen3-1.7B is robust where Llama-3.1-8B collapses, and we further show the instability is termination failure rather than factual drift. A related test-time-scaling line attributes temperature-flattening in reasoning models to internal deliberation; our robust models flatten with thinking mode off (a thinking-on spot-check agrees), so deliberation is not the mechanism here.

3 Experimental setup

One pipeline, everything pinned: llama.cpp at a fixed commit (CUDA build), one provider’s imatrix GGUF quantizations with logged SHA256 hashes, official chat templates rendered client-side with the exact prompt string hashed into each record, the native completion API, a pinned sampler chain order (temperature first; the llama.cpp default of temperature last is included as an ablation arm for top- p and min- p), per-record derived seeds, a deterministic greedy anchor ($K=1$), and $K=3$ repetitions elsewhere.

Tasks. GSM8K (0-shot chain-of-thought, the Llama-eval prompt, numeric exact match) and MMLU-Pro (0-shot chain-of-thought, letter exact match), $N=50$ items each, with parsers faithful to lm-eval-harness. The parse path (strict final-answer sentence, flexible fallback, or failure) is logged per record. Parse failure grades as incorrect *and* is reported as a metric; Section 4.3 shows it is a result, not a nuisance. One caveat stated up front: the frozen $N=50$ MMLU-Pro subset is head-of-split, and the split is sorted by category, so the subset is single-category (business). All within-study contrasts hold items fixed; a stratified $N=50$ replication (3–4 items from each of the 14 categories) reproduces the headline result (Section 4.1, Appendix C).

Models and quantization. Llama-3.1-8B, Mistral-7B-v0.3, Qwen2.5-7B, Qwen3-1.7B/4B/8B (thinking off; a thinking-on spot-check in Appendix F), and Gemma-3-12B, all instruction-tuned. Quantization levels Q8_0, Q6_K, Q4_K_M, Q3_K_M, plus a BF16 anchor on Qwen3-4B that matches Q8 within noise.

Inference. Item-clustered nonparametric bootstrap ($B=2000$), paired where the comparison shares items.

Hardware. Generation ran under one pinned llama.cpp commit on two setups: RTX 2080 Ti (11 GB) nodes of a university Slurm cluster (17 of the 21 main-matrix cells) and a single RTX 5070 Ti (16 GB) workstation (the four VRAM-heavy cells and every follow-up run), with per-architecture CUDA builds of the same commit. About 202,000 graded generations in total. Full reproducibility details in Appendix I.

Table 1: Pure-temperature accuracy drop $T0.7 \rightarrow T1.3$ (pp, 95% item-clustered bootstrap CIs), pooled over quantization levels.

Model	GSM8K	MMLU-Pro
Llama-3.1-8B	+34.0 [28.5, 39.7]	+40.8 [31.8, 50.3]
Mistral-7B-v0.3	+3.5 [-1.2, 8.5]	+8.0 [3.5, 13.2]
Qwen2.5-7B	+4.0 [1.2, 7.3]	+4.8 [1.3, 8.8]
Gemma-3-12B	+2.0 [-0.0, 4.8]	+3.7 [0.2, 7.7]
Qwen3-8B	+0.5 [-1.5, 2.6]	+1.0 [-2.4, 4.8]
Qwen3-4B	+2.1 [0.4, 3.9]	+0.7 [-1.0, 2.5]
Qwen3-1.7B	-0.3 [-3.8, 3.0]	+0.7 [-3.4, 5.0]

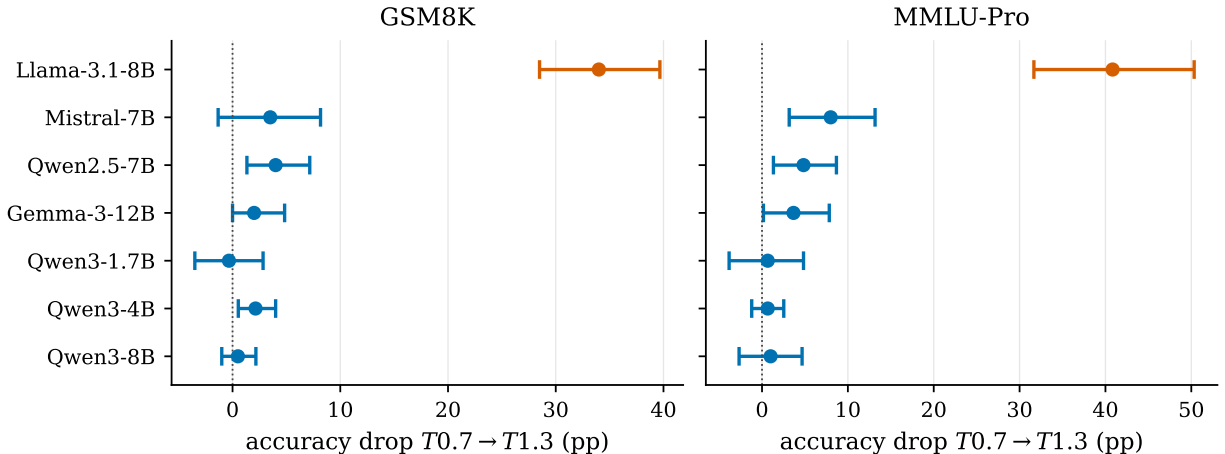


Figure 1: The pure-temperature collapse is model-bound: accuracy drop $T0.7 \rightarrow T1.3$, points with 95% item-clustered bootstrap CIs.

4 Results

4.1 The collapse is model-bound

Table 1 and Figure 1 give the accuracy drop under plain temperature sampling from $T=0.7$ to $T=1.3$, pooled over quantization levels.

The effect is a gradient, not a binary: the Qwen3 family is a tight null, Qwen2.5, Gemma, and Mistral show small but real drops, and Llama is catastrophic. Four controls rule out the obvious confounds. It is not accuracy headroom: Qwen3-1.7B (74% on GSM8K) sits in Llama’s accuracy regime (78%) with the opposite temperature behavior. It is not model age: Mistral-7B-v0.3 predates Llama-3.1 and is robust. It is not the item subset: the stratified MMLU-Pro replication gives Llama +43.3 pp [31.3, 55.3], Qwen2.5 +6.0, and Gemma +4.5. And it is not instruction tuning alone, since every model here is instruction-tuned [4].

4.2 Sampler choice barely matters before the cliff

At $T=1.3$ the maximum minus minimum accuracy across the seven stochastic sampler arms is 41.8 pp (MMLU-Pro) and 36.7 pp (GSM8K) for Llama, and 1.3 to 10.0 pp for every other model

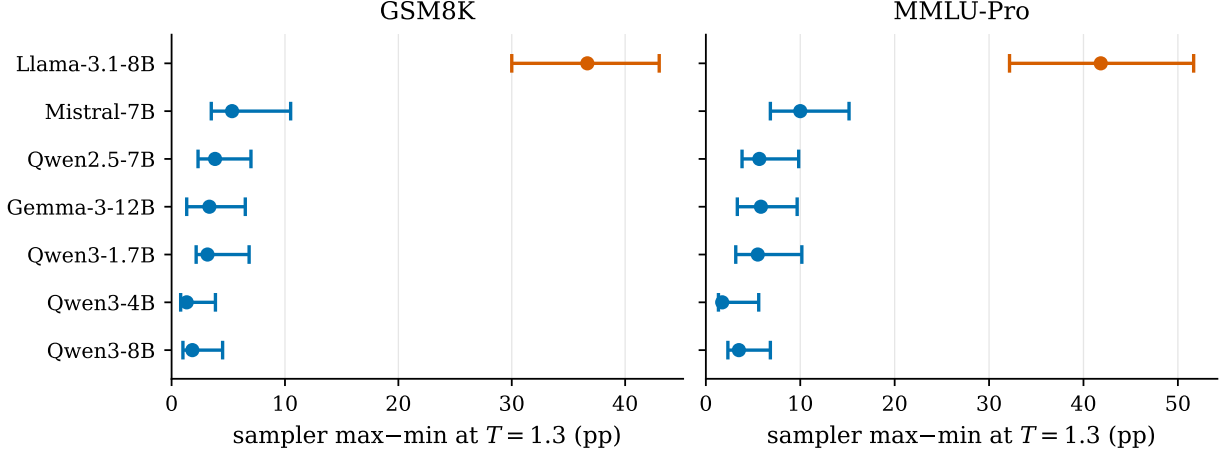


Figure 2: Sampler max-min accuracy spread at $T=1.3$ per model (95% CIs). The statistic is biased upward, so robust-model spreads are upper bounds.

(Figure 2). The max-minus-min statistic is biased upward under sampling noise, so the small spreads are upper bounds; the true differences on robust models are smaller. Chain order (temperature first versus last) is a no-op on robust models ($\pm 1-2$ pp) and matters only where everything matters: on Llama at $T=1.3$, temperature-last truncation is more protective (top- p 64% $\rightarrow$ 73%, min- p 72% $\rightarrow$ 76% on GSM8K), as expected, because truncating before temperature scaling fixes the surviving token set at its low-temperature size.

4.3 The collapse is degeneration, not wrong answers

We decompose accuracy by parse path (Figure 3). What collapses is termination: at $T=1.3$, 53.8% (GSM8K) and 89.2% (MMLU-Pro) of Llama generations derail into token-level noise and run to the token cap. Among generations that do produce a well-formed final answer, Llama scores 97.3% on GSM8K at $T=1.3$. Survival is selective (harder items derail first: greedy accuracy on the surviving items is 95.8%, against 76.0% over all items), so the properly normalized statement is that, *conditional on surviving, accuracy at $T=1.3$ matches greedy decoding on the same items* (97.3 vs. 95.8% on GSM8K; 76.9 vs. 78.8% on MMLU-Pro). Temperature’s entire marginal effect is on whether the generation survives, not on which answer it picks.

The two tasks disguise one failure mode differently. Degenerate text almost always contains a number, so the GSM8K flexible last-number fallback grades it as a wrong answer; the MMLU-Pro letter parser reports it as a parse failure. Temperature does not make a model choose worse answers; it makes a fragile model fail to finish.

4.4 The cliff map: robustness is cliff location

We extend the temperature axis to $T \in \{1.3, 1.5, 1.7, 2.0\}$ on three models spanning the gradient (Q8, both tasks, $N=50$, $K=3$; Figure 4). Under plain temperature on GSM8K the three models form a staircase, one rung apart: Llama falls 45 $\rightarrow$ 9 $\rightarrow$ 1 $\rightarrow$ 0%, Qwen2.5 falls 89 $\rightarrow$ 69 $\rightarrow$ 23 $\rightarrow$ 2%, and Gemma falls 89 $\rightarrow$ 77 $\rightarrow$ 49 $\rightarrow$ 8%. The drops against each model’s own $T=0.7$ anchor order the staircase at every rung: at $T=1.5$ Llama (+68.0 pp [58.0, 76.7]) is disjoint from both Qwen2.5 (+24.0 [15.3, 33.3]) and Gemma (+14.7 [8.7, 21.3]), whose intervals still touch; by $T=1.7$

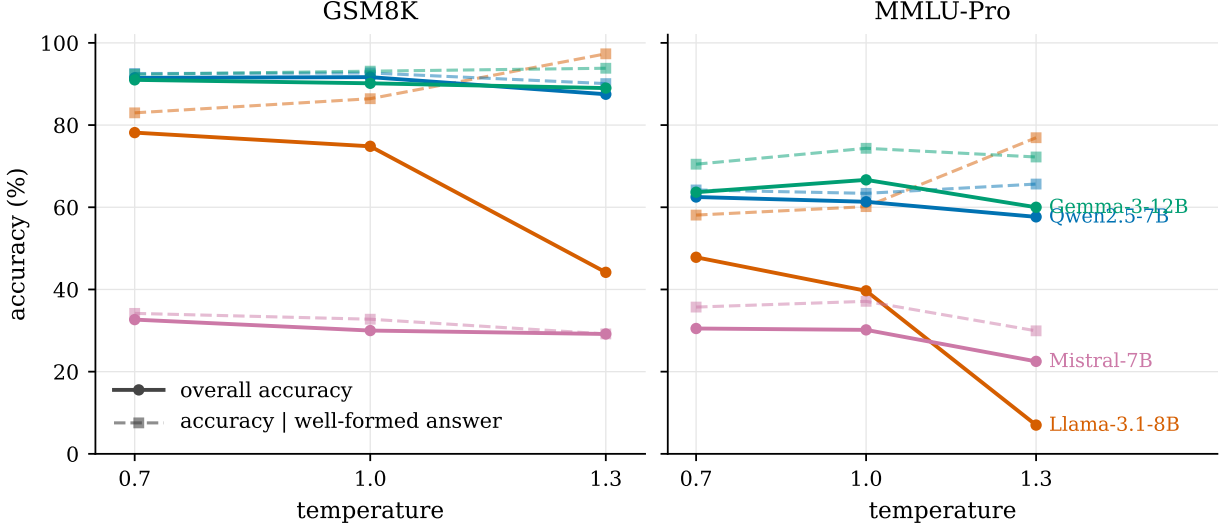


Figure 3: Overall accuracy (solid) versus accuracy among well-formed, strict-parsed answers (dashed) under plain temperature sampling. The dashed lines are flat: the collapse is degeneration, not answer selection.

all three are pairwise disjoint (Qwen2.5 +70.0 [62.0, 77.3] versus Gemma +42.7 [34.7, 50.7]). Every model has a cliff; what a robust model owns is the cliff’s location, roughly 0.2 in temperature per step of the gradient. Within each model the cliff sits earlier on MMLU-Pro (Llama is at 3% already at $T=1.3$; Qwen2.5 falls $57 \rightarrow 25 \rightarrow 4 \rightarrow 0\%$; Gemma falls $60 \rightarrow 24 \rightarrow 6 \rightarrow 0\%$), the task with roughly twice the generation length, which is what a compounding per-token derail risk predicts. The longer task also compresses the staircase: Qwen2.5 and Gemma land on the same MMLU-Pro rung ($T=1.5$) while remaining one rung apart on GSM8K.

Truncation shifts the cliff right in order of tail-cutting strength. Top- p buys about 0.2 in temperature. Min- p holds 70% at $T=1.5$ on Llama GSM8K where plain temperature is at 9%, 85–87% on Qwen2.5 and Gemma GSM8K out to $T=2.0$, and 53 to 59% on Qwen MMLU-Pro. Top- $n\sigma$ is flat to $T=2.0$ on all three models (75 to 93% on GSM8K; 56 to 66% on Qwen MMLU-Pro), consistent with its construction: the surviving token set is temperature-invariant. The strict-parse rate tracks accuracy almost one-to-one along the whole ladder, so the cliff *is* the degeneration cliff. This reconciles the sampler papers with our null result: min- p and top- $n\sigma$ validate at $T \geq 1.5$ to 3, past every model’s deployment range. There the gains are real and ordered top- $n\sigma \gg$ min- $p \gg$ top- p (the $T=2\text{--}3$ probe in Appendix D), but at $T \leq 1.3$ on a robust model there is nothing left for a sampler to fix.

4.5 Quantization and self-consistency

Greedy accuracy from Q8_0 to Q3_K_M (pooled tasks) is null on every model of 7B parameters or more (Gemma −3.0 [−9, +3]) and −9 pp on Qwen3-1.7B: the effect grows as models shrink, and nowhere does it reorder samplers (Figure 6). Self-consistency at $T=1.3$: robust models emit three distinct token sequences across $K=3$ repetitions yet converge to 1.1 to 1.4 distinct parsed answers per item; Llama and Mistral diverge to 2.2 to 2.5. Robustness is convergence in answer space, not a peaked next-token distribution.

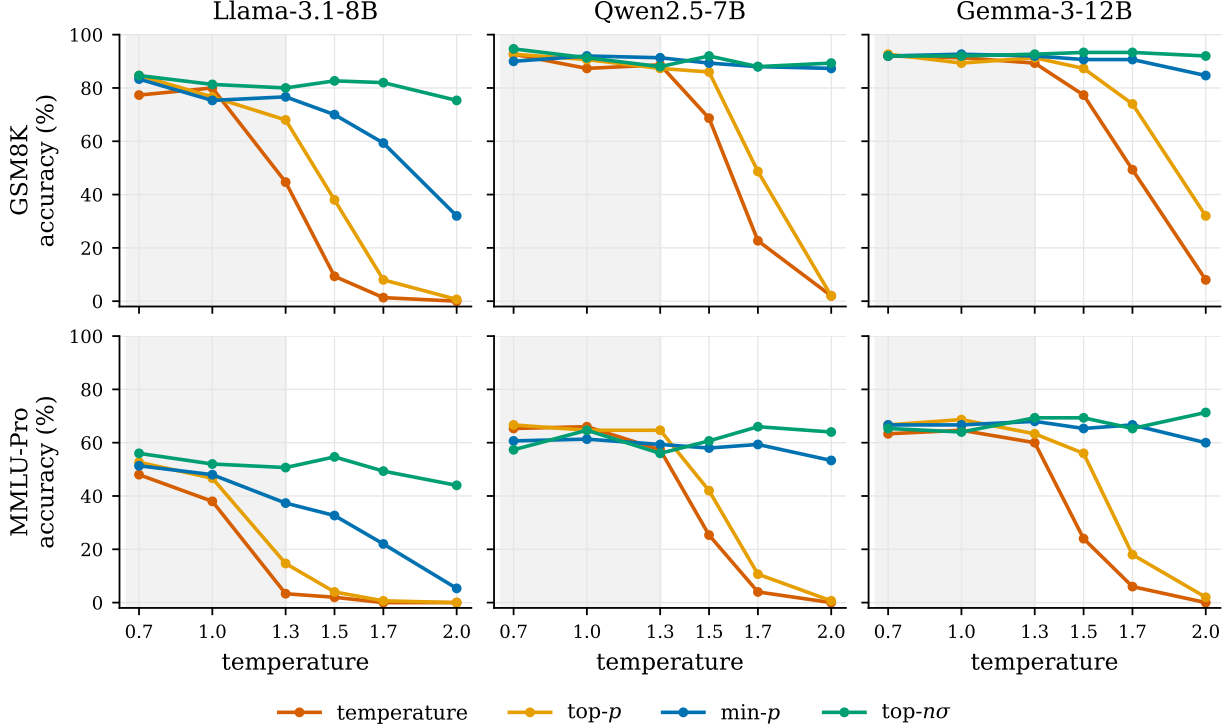


Figure 4: The cliff map. Accuracy versus temperature per sampler (anchors $T \leq 1.0$ from the main matrix, Q8). The cliff location, not its existence, is the model property; truncation shifts the cliff right by tail-cutting strength.

4.6 Toward a mechanism: per-token uncertainty ranks with fragility

If the collapse is a derailment process, fragile models should carry more per-token uncertainty before any derailment happens. We probe this directly: for every model (Q8) we decode the frozen 100 prompts greedily and record the raw top-20 next-token probabilities at each position, summarized per prompt as an entropy lower bound, the top-1 probability, the top-1 minus top-2 margin, and the fraction of low-confidence positions (top-1 < 0.5). Greedy decoding holds the trajectory at the model’s modal path, so the measurement is not confounded by sampled derailments.

Mean greedy-path entropy ranks with the observed temperature drop at Spearman $\rho = +0.96$ across the seven models (flat-position fraction +0.96; top-1 margin −0.96; Figure 5). The ranking is informative and so is its failure: Llama-3.1 (entropy 0.33 nats) and Mistral (0.32) are nearly tied on every top-20 summary, including the mass beyond the top 20 (1.0×10^{-3} both), yet their drops differ five-fold (40.8 versus 8.0 pp).

Generation lengths say where the missing mechanism is not. If models derailed and recovered, terminated generations at $T=1.3$ would carry the extra tokens of the detour; instead, on every robust model the terminated-length distribution is essentially unchanged from $T=0.7$ (90th-percentile ratio 0.85 to 1.19 across models and tasks; Table 4). Llama’s surviving generations at $T=1.3$ are *shorter* than at $T=0.7$ (90th percentile $\times 0.73$ on GSM8K, $\times 0.55$ on MMLU-Pro): exactly the censoring an absorbing failure with a compounding per-token hazard produces, where the generations that survive are the ones short enough to finish before the first bad draw. Derailment is an absorbing state in every model.

A direct estimate of the hazard itself closes the remaining door. Re-decoding the frozen prompts

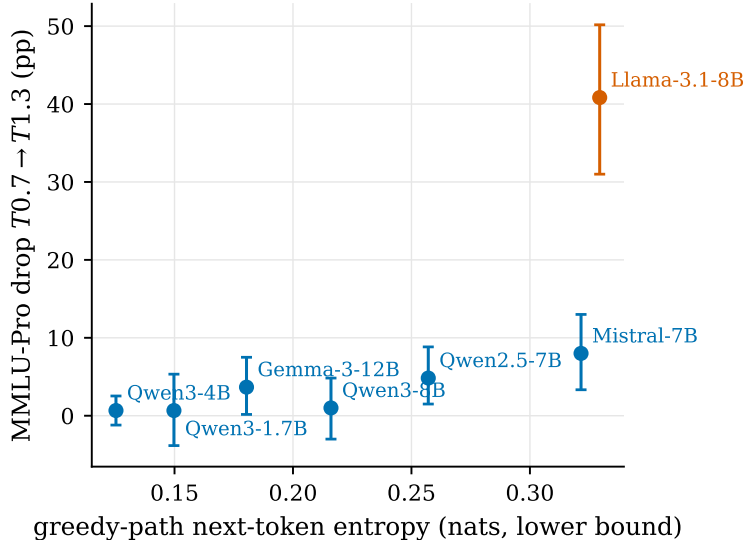


Figure 5: Per-token uncertainty on the model’s own greedy path versus the temperature drop (MMLU-Pro, 95% CIs). The rank correlation is $\rho = +0.96$; the Llama–Mistral gap at equal entropy shows flatness alone does not set the magnitude.

and storing the raw top-20 arrays at every position, we compute the temperature-scaled probability of drawing a token from beyond the top 20 at $T=1.3$: 0.0033 per position for Llama versus 0.0034 for Mistral (about 0.85 expected tail draws per 256 positions for both; across all seven models this hazard ranks with the drop at the same $\rho = +0.96$). Both models take the first off-distribution token equally often; only Llama spirals on it. A single absorbed token leaves no length footprint, so this is consistent with the unchanged terminated lengths above. The fragility is therefore not in the next-token distribution at all, scaled or unscaled: it is the model’s instability to a single off-distribution token in its own context.

A perturbation experiment measures that instability directly. For each model and frozen prompt we greedily-decode a 48-token prefix, force the model’s own rank-20 candidate at that position (median raw probability 2×10^{-9} : a genuine tail draw), and continue greedily for 256 tokens against a control continuation of the unedited prefix. The forced token is absorbed on the surface: continuations stay locally coherent, and Llama’s reach the token cap no more often than its controls (14.0% both, GSM8K). The destabilization is underneath. Over the 32 positions after the injection, Llama’s beyond-top-20 escape hazard at $T=1.3$ rises to 2.3 times its baseline (+0.0042 per position); Mistral’s rises 1.35 times (+0.0012), and no robust model gains more than +0.0014. The per-model destabilization ranks with the observed temperature drop at $\rho = +0.86$ (entropy delta: +0.82). This is the compounding loop measured directly: one off-distribution token in context multiplies the probability of drawing the next one, and what separates the fragile model is the multiplier, not the base rate (Appendix H consolidates all per-model probe numbers). With seven models these are consistency checks, not a confirmed causal account, and we report them as such.

5 Discussion

For practitioners. Pick the model, not the sampler. Sampler tuning is insurance against a failure mode that current post-training has largely removed at deployment temperatures.

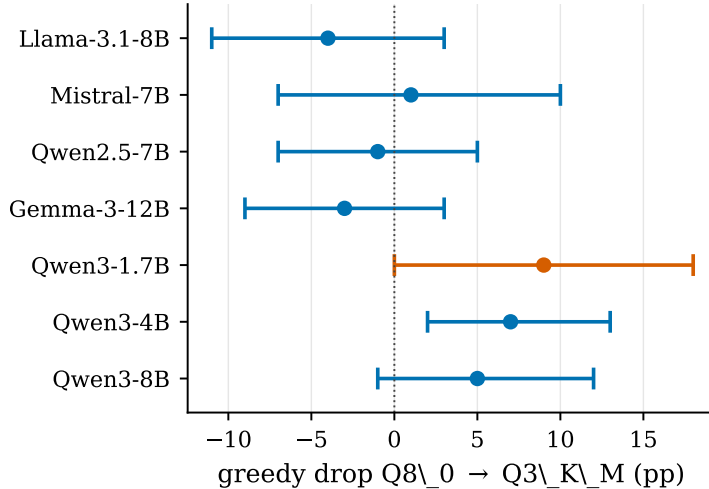


Figure 6: Greedy accuracy drop $Q8_0 \rightarrow Q3_K_M$ per model (95% CIs), pooled tasks.

For the sampler literature. Truncation samplers solve a real but receding problem. Their standard evaluation regime ($T \geq 2$) measures the size of the safety margin, not deployment gains. Top- $n\sigma$'s temperature invariance does exactly what it promises, but one has to stand past the cliff to see it.

For the robustness line. The variable is the model, not instruction tuning. What changed between Llama-3.1 and the 2025 cohort (post-training recipe, data, RLHF) is an open question; we contribute the behavioral signature a mechanism study should explain: termination failure with intact answer selection, triggered by off-distribution tokens that arrive at the same per-token rate in robust models and are absorbed only by them.

Why degeneration matters beyond benchmarks. An answer-selection failure and a derailment failure have different production consequences (silent errors versus detectable garbage) and different fixes (output validation versus sampler guard rails).

6 Limitations

$N=50$ items per task (CIs are shown throughout; the headline effects are 5 to 10 times their CI widths). 0-shot prompting, English, two tasks. The MMLU-Pro head subset is single-category; the stratified replication is in Appendix C. One fragile model: we characterize Llama-3.1, not the class of fragile models, and whether fragility is lineage-specific is untested. Mistral is near floor on GSM8K (32.7% at $T=0.7$), which limits its collapse-detection power there (its MMLU-Pro CI excludes zero). The max-minus-min spread is biased upward, which works against our claim. Accuracy conditional on strict parsing has item survivorship, quantified and normalized in Section 4.3. llama.cpp k-quants only, one quantization recipe by design. Thinking mode off by default (the thinking-on-spot-check in Appendix F agrees).

7 Conclusion

Across seven models, five samplers, four quantization levels, and two reasoning tasks, we find that the choice of sampler is a second-order effect at deployment temperatures: on robust models the

max-minus-min accuracy spread across samplers is 1.3 to 10.0 points at $T=1.3$, while a single model, Llama-3.1-8B, loses 34 to 41 points under pure temperature on the same grid. The collapse is not an answer-selection failure but a termination failure, every model has such a cliff somewhere above its deployment range, and what the literature calls robustness is the location of that cliff; truncation samplers shift it rightward but pay off only at temperatures nobody deploys. Per-token entropy on the greedy decoding path ranks models by fragility (Spearman $\rho = +0.96$ across the seven models), and a single forced tail token multiplies the fragile model’s subsequent escape hazard 2.3-fold where robust models absorb it, which together suggest cheap screening before deployment rather than sampler tuning after it. The open question our data sharpen but cannot answer is what, in post-training, moves the cliff.

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A Full sampler–temperature grids

Accuracy by sampler and temperature, pooled over quantization levels (Tables 2 and 3).

B Degeneration decomposition

Table 5 decomposes accuracy under plain temperature sampling by parse path: the strict final-answer-sentence rate, the parse-failure rate, the fraction of generations that ran to the token cap (including server-recovered degenerate outputs), and accuracy conditional on a strict parse. Two representative degenerate outputs (Llama-3.1-8B, Q8, plain temperature, $T=1.3$, MMLU-Pro; both ran to the 1024-token cap):

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[opens] To solve this problem, we'll use the gross profit method of inventory evaluation.
Step 1: Calculate the gross profit earned during the year...
[ends] H sieve Con Palette sediment conduit France called coaching OMAX may PHYS
today Enhanced vertices XOR/t and registration ordinances efforts yesterday Michael

[opens] To determine the correct type of research methods, we need to understand
the characteristics of each option: A. Non-probability: This method involves selecting...
[ends] entageeba lever modeling ign measures inherits cones quantum auditingippines
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The pattern is consistent: a coherent opening, a mid-generation derailment, then token noise until the cap. The model does not produce a confident wrong answer; it never gets to an answer.

Table 4 gives the length-censoring evidence used in Section 4.6: run-to-cap rates and the 90th percentile of completion length among *terminated* generations at $T=0.7$ versus $T=1.3$. Robust models are unchanged (ratio 0.85 to 1.19); Llama’s surviving generations shorten ($\times 0.73$ GSM8K, $\times 0.55$ MMLU-Pro), so derailment is absorbing and survival is decided by whether the generation finishes before the first bad draw.

C Stratified MMLU-Pro replication

The frozen $N=50$ MMLU-Pro head subset is single-category (business) because the test split is sorted by category. Table 6 repeats the pure-temperature axis (Q8, $K=3$) on a deterministic stratified subset with 3–4 items from each of the 14 categories. The collapse ordering reproduces.

D High-temperature probe ($T=2-3$)

A small probe ($N=15$, $K=3$, Q8) in the regime where min- p and top- $n\sigma$ report their largest gains (Table 7). The sampler ordering is top- $n\sigma \gg \text{min-}p \gg \text{top-}p \approx \text{temperature}$; the zero cells are entirely degeneration-to-cap, and even the most robust model collapses under plain temperature.

E Temperature-first versus temperature-last

Both chain orders for the two probability-space truncators (Table 8).

F Thinking-mode spot-check

The matrix runs Qwen3 with thinking off. A confirmatory spot-check (Qwen3-8B, Q8, thinking on, 450 generations, $N=25$, $K=2$) shows the same temperature robustness: under plain temperature, GSM8K 94% at $T=0.7$ versus 96% at $T=1.3$, and MMLU-Pro 56% versus 46% (small sample), with min- p flat on both tasks. The robustness is not an artifact of suppressing the reasoning mode.

G Self-consistency

Mean distinct parsed answers per item across the $K=3$ repetitions under plain temperature (Table 9). Robust models sample diverse token sequences that converge on the same answer; a parse failure counts as one distinct answer, so the statistic is a divergence proxy.

H Mechanism probes

Table 10 consolidates the per-model mechanism evidence of Section 4.6: greedy-path entropy lower bound and flat-position fraction ($\text{top-1} < 0.5$) from the summary probe; the temperature-scaled beyond-top-20 escape hazard per position at $T=1.3$ from the raw top-20 arrays; and the perturbation deltas (post-injection minus control over 32 positions, escape hazard and entropy) from the forced-token experiment. Escape columns are $\times 10^{-3}$. Every static column ties Llama-3.1 with Mistral; only the perturbation response separates them.

I Reproducibility

- **Inference stack.** llama.cpp at commit 5aba5364 (server build 9456), native `/completion` endpoint, CUDA builds of the same commit per GPU architecture (sm_75 for RTX 2080 Ti, sm_120 for RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special tokens, and each record stores the SHA-256 of the exact rendered prompt.
- **Quantizations.** Every quantized checkpoint is a Bartowski imatrix GGUF (one provider, one recipe, so the quantization axis is not confounded by recipe). SHA-256 hashes of all 29 model files are pinned in the repository (DOWNLOADS.md and per-file .meta.json sidecars).
- **Decoding.** Exactly one truncation parameter per request; the sampler chain order is pinned per arm and recorded in every record. Sampler values: top- p 0.95, top- k 40, min- p 0.05, top- $n\sigma$ 1.0 (each method’s recommended value).
- **Seeding.** Every generation’s seed is derived deterministically from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Greedy decoding is the deterministic $T=0$ anchor with $K=1$.

- **Tasks.** GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items; stratified replication in Appendix C). Parsers mirror lm-eval-harness; parse failures grade as incorrect and are reported.
- **Records.** Each generation is one JSONL record carrying the full identity tuple, the request parameters, the raw output, the parse path, and throughput covariates; runs are resumable by record id. Batched serving (≤ 8 slots) is seed-logged but not bit-exact.
- **Code and data.** [\[TODO: anonymized repository link for submission.\]](#)

Table 2: GSM8K accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	78.0	–	–
	temperature	78.2	74.8	44.2
	top- k	79.8	75.2	58.7
	top- p	81.5	78.0	63.8
	min- p	81.2	78.0	72.0
	top- $n\sigma$	78.8	80.7	80.8
	top- p (temp-last)	80.8	78.7	73.0
	min- p (temp-last)	78.7	81.3	76.2
Mistral-7B	greedy ($T=0$)	31.0	–	–
	temperature	32.7	30.0	29.2
	top- k	31.8	31.7	29.8
	top- p	28.2	31.0	34.2
	min- p	32.3	28.3	28.8
	top- $n\sigma$	29.5	30.7	30.2
	top- p (temp-last)	29.5	31.2	29.5
	min- p (temp-last)	29.5	32.0	30.5
Qwen2.5-7B	greedy ($T=0$)	93.5	–	–
	temperature	91.5	91.7	87.5
	top- k	90.2	88.5	88.7
	top- p	92.2	91.2	90.3
	min- p	90.8	90.5	91.0
	top- $n\sigma$	92.7	91.2	90.7
	top- p (temp-last)	90.8	91.3	90.3
	min- p (temp-last)	92.0	90.7	91.3
Gemma-3-12B	greedy ($T=0$)	92.0	–	–
	temperature	91.0	90.2	89.0
	top- k	91.8	91.7	90.7
	top- p	91.5	90.3	90.7
	min- p	91.5	91.3	92.3
	top- $n\sigma$	91.3	91.5	91.0
	top- p (temp-last)	90.2	91.8	91.5
	min- p (temp-last)	91.2	90.8	90.3
Qwen3-1.7B	greedy ($T=0$)	75.5	–	–
	temperature	73.3	73.7	73.7
	top- k	72.8	72.7	73.3
	top- p	73.8	75.0	72.7
	min- p	72.2	71.7	74.0
	top- $n\sigma$	74.3	73.5	72.3
	top- p (temp-last)	73.0	72.0	71.3
	min- p (temp-last)	74.7	73.3	74.5
Qwen3-4B	greedy ($T=0$)	95.2	–	–
	temperature	95.2	94.0	93.1
	top- k	94.0	94.5	94.1
	top- p	95.5	95.5	93.5
	min- p	95.1	94.7	92.9
	top- $n\sigma$	94.8	93.9	93.9
	top- p (temp-last)	94.8	95.2	94.3
	min- p (temp-last)	94.5	93.9	93.9
Qwen3-8B	greedy ($T=0$)	94.5	–	–
	temperature	95.0	94.2	94.5
	top- k	94.8	94.8	93.5
	top- p	94.0	94.3	93.7
	min- p	94.8	95.3	94.7
	top- $n\sigma$	96.3	94.2	94.3

Table 3: MMLU-Pro accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	54.0	–	–
	temperature	47.8	39.7	7.0
	top- k	51.2	43.3	26.8
	top- p	51.3	46.0	16.0
	min- p	51.2	50.8	39.5
	top- $n\sigma$	51.0	50.5	48.8
	top- p (temp-last)	50.3	46.7	37.5
	min- p (temp-last)	51.2	50.2	43.5
Mistral-7B	greedy ($T=0$)	35.5	–	–
	temperature	30.5	30.2	22.5
	top- k	32.5	32.3	28.3
	top- p	30.2	29.5	25.3
	min- p	33.8	30.7	31.0
	top- $n\sigma$	32.3	31.5	31.5
	top- p (temp-last)	29.2	30.2	32.5
	min- p (temp-last)	33.2	33.3	28.7
Qwen2.5-7B	greedy ($T=0$)	64.0	–	–
	temperature	62.5	61.3	57.7
	top- k	62.5	61.7	57.0
	top- p	65.5	62.8	62.7
	min- p	61.3	63.5	59.7
	top- $n\sigma$	62.0	63.5	61.0
	top- p (temp-last)	63.7	62.7	61.3
	min- p (temp-last)	64.2	63.0	62.3
Gemma-3-12B	greedy ($T=0$)	61.0	–	–
	temperature	63.7	66.7	60.0
	top- k	64.3	62.8	63.7
	top- p	65.7	66.7	63.8
	min- p	65.3	67.0	65.5
	top- $n\sigma$	65.2	65.8	65.8
	top- p (temp-last)	65.8	65.0	65.2
	min- p (temp-last)	65.3	63.7	63.8
Qwen3-1.7B	greedy ($T=0$)	45.5	–	–
	temperature	46.0	45.7	45.3
	top- k	47.5	46.5	45.0
	top- p	47.2	47.0	47.2
	min- p	47.0	47.7	46.2
	top- $n\sigma$	46.8	47.2	46.2
	top- p (temp-last)	47.7	46.8	43.5
	min- p (temp-last)	45.7	46.2	49.0
Qwen3-4B	greedy ($T=0$)	61.6	–	–
	temperature	61.9	62.5	61.2
	top- k	63.5	61.1	61.2
	top- p	62.8	61.3	62.8
	min- p	60.9	62.7	62.9
	top- $n\sigma$	62.4	62.4	61.5
	top- p (temp-last)	62.9	62.7	61.7
	min- p (temp-last)	61.3	63.6	61.5
Qwen3-8B	greedy ($T=0$)	60.5	–	–
	temperature	60.5	59.0	59.5
	top- k	60.2	61.3	57.0
	top- p	60.5	61.3	60.5
	min- p	61.0	59.2	59.7
	top- $n\sigma$	60.0	58.8	58.7

Table 4: Terminated-length censoring under plain temperature (pooled over quantization). “At cap” includes server-recovered degenerate streams.

Model	Task	at cap (%)		p90 length		ratio
		$T=0.7$	$T=1.3$	$T=0.7$	$T=1.3$	
Llama-3.1-8B	GSM8K	5.2	53.8	318	232	$0.73\times$
	MMLU-Pro	11.0	89.2	633	345	$0.55\times$
Mistral-7B	GSM8K	3.5	8.3	367	378	$1.03\times$
	MMLU-Pro	1.0	12.5	628	750	$1.19\times$
Qwen2.5-7B	GSM8K	0.2	1.7	341	361	$1.06\times$
	MMLU-Pro	2.5	10.2	771	732	$0.95\times$
Gemma-3-12B	GSM8K	2.0	5.3	363	344	$0.95\times$
	MMLU-Pro	9.8	17.2	753	763	$1.01\times$
Qwen3-1.7B	GSM8K	2.0	2.2	324	339	$1.05\times$
	MMLU-Pro	15.8	16.3	779	806	$1.03\times$
Qwen3-4B	GSM8K	0.9	0.8	353	361	$1.02\times$
	MMLU-Pro	11.9	12.4	778	731	$0.94\times$
Qwen3-8B	GSM8K	2.3	2.3	379	389	$1.03\times$
	MMLU-Pro	12.3	16.0	850	724	$0.85\times$

Table 5: Decomposition of the temperature effect (plain temperature, pooled over quantization; %).

Model	Task	T	acc	strict	failed	at cap	acc strict
Llama-3.1-8B	GSM8K	0.7	78.2	93.0	0.0	5.2	83.0
	GSM8K	1.0	74.8	84.7	0.0	7.8	86.4
	GSM8K	1.3	44.2	43.8	0.5	53.8	97.3
	MMLU-Pro	0.7	47.8	82.3	15.3	11.0	58.1
	MMLU-Pro	1.0	39.7	64.8	29.0	23.7	60.2
	MMLU-Pro	1.3	7.0	8.7	84.5	89.2	76.9
Mistral-7B	GSM8K	0.7	32.7	72.2	0.3	3.5	34.2
	GSM8K	1.0	30.0	67.7	0.0	4.3	32.8
	GSM8K	1.3	29.2	62.8	0.0	8.3	29.2
	MMLU-Pro	0.7	30.5	84.0	12.8	1.0	35.7
	MMLU-Pro	1.0	30.2	80.8	17.0	1.0	37.1
	MMLU-Pro	1.3	22.5	73.5	21.2	12.5	29.9
Qwen2.5-7B	GSM8K	0.7	91.5	98.3	0.0	0.2	92.5
	GSM8K	1.0	91.7	95.7	0.0	0.5	92.7
	GSM8K	1.3	87.5	92.5	0.0	1.7	90.1
	MMLU-Pro	0.7	62.5	97.3	2.3	2.5	64.2
	MMLU-Pro	1.0	61.3	96.5	2.5	3.0	63.4
	MMLU-Pro	1.3	57.7	87.3	10.5	10.2	65.6
Gemma-3-12B	GSM8K	0.7	91.0	98.0	0.0	2.0	92.3
	GSM8K	1.0	90.2	96.8	0.0	3.3	93.1
	GSM8K	1.3	89.0	94.5	0.0	5.3	93.8
	MMLU-Pro	0.7	63.7	90.3	9.0	9.8	70.5
	MMLU-Pro	1.0	66.7	89.7	8.2	10.3	74.3
	MMLU-Pro	1.3	60.0	82.8	14.5	17.2	72.2
Qwen3-1.7B	GSM8K	0.7	73.3	20.3	0.0	2.0	73.8
	GSM8K	1.0	73.7	20.0	0.0	0.7	71.7
	GSM8K	1.3	73.7	19.0	0.0	2.2	71.1
	MMLU-Pro	0.7	46.0	83.7	13.3	15.8	54.8
	MMLU-Pro	1.0	45.7	82.2	15.5	16.3	55.4
	MMLU-Pro	1.3	45.3	83.2	13.5	16.3	54.5
Qwen3-4B	GSM8K	0.7	95.2	94.3	0.0	0.9	95.3
	GSM8K	1.0	94.0	92.3	0.0	0.8	94.8
	GSM8K	1.3	93.1	95.6	0.0	0.8	93.7
	MMLU-Pro	0.7	61.9	87.2	12.5	11.9	70.9
	MMLU-Pro	1.0	62.5	87.3	12.0	10.8	71.6
	MMLU-Pro	1.3	61.2	87.5	11.1	12.4	70.0
Qwen3-8B	GSM8K	0.7	95.0	48.2	0.0	2.3	98.3
	GSM8K	1.0	94.2	49.2	0.0	2.3	98.6
	GSM8K	1.3	94.5	52.2	0.0	2.3	98.1
	MMLU-Pro	0.7	60.5	82.5	15.2	12.3	72.3
	MMLU-Pro	1.0	59.0	80.5	15.5	14.8	72.3
	MMLU-Pro	1.3	59.5	79.8	16.3	16.0	73.5

Table 6: Stratified versus head-of-split (business) MMLU-Pro subset; plain temperature accuracy (%) at Q8_0.

Model	Subset	$T=0.7$	$T=1.0$	$T=1.3$	drop
Llama-3.1-8B	stratified	44.7	38.7	1.3	+43.3 pp
Llama-3.1-8B	business (head-50)	48.0	38.0	10.0	+38.0 pp
Qwen2.5-7B	stratified	54.7	53.3	48.7	+6.0 pp
Qwen2.5-7B	business (head-50)	65.3	66.0	58.0	+7.3 pp
Gemma-3-12B	stratified	64.0	62.7	56.7	+7.3 pp
Gemma-3-12B	business (head-50)	63.3	64.7	60.0	+3.3 pp

Table 7: Accuracy (%) at $T=2$ and $T=3$ (45 observations per cell).

Task	Sampler	Llama-3.1-8B		Gemma-3-12B	
		$T=2$	$T=3$	$T=2$	$T=3$
GSM8K	temperature	0	0	16	0
	top- p	0	0	27	0
	min- p	33	0	78	24
	top- $n\sigma$	71	80	87	89
MMLU-Pro	temperature	0	0	0	0
	top- p	0	0	0	0
	min- p	0	0	58	0
	top- $n\sigma$	47	51	69	62

Table 8: Accuracy (%) with temperature applied before (left) versus after (right) truncation, pooled over quantization.

Model	Method	temperature-first			temperature-last		
		0.7	1.0	1.3	0.7	1.0	1.3
<i>GSM8K</i>							
Llama-3.1-8B	top- <i>p</i>	81.5	78.0	63.8	80.8	78.7	73.0
Llama-3.1-8B	min- <i>p</i>	81.2	78.0	72.0	78.7	81.3	76.2
Mistral-7B	top- <i>p</i>	28.2	31.0	34.2	29.5	31.2	29.1
Mistral-7B	min- <i>p</i>	32.3	28.3	28.8	29.5	32.0	30.5
Qwen2.5-7B	top- <i>p</i>	92.2	91.2	90.3	90.8	91.3	90.3
Qwen2.5-7B	min- <i>p</i>	90.8	90.5	91.0	92.0	90.7	91.3
Gemma-3-12B	top- <i>p</i>	91.5	90.3	90.7	90.2	91.8	91.5
Gemma-3-12B	min- <i>p</i>	91.5	91.3	92.3	91.2	90.8	90.3
Qwen3-1.7B	top- <i>p</i>	73.8	75.0	72.7	73.0	72.0	71.3
Qwen3-1.7B	min- <i>p</i>	72.2	71.7	74.0	74.7	73.3	74.5
Qwen3-4B	top- <i>p</i>	95.5	95.5	93.5	94.8	95.2	94.3
Qwen3-4B	min- <i>p</i>	95.1	94.7	92.9	94.5	93.9	93.9
Qwen3-8B	top- <i>p</i>	94.0	94.3	93.7	93.8	94.0	94.0
Qwen3-8B	min- <i>p</i>	94.8	95.3	94.7	96.2	94.7	95.3
<i>MMLU-Pro</i>							
Llama-3.1-8B	top- <i>p</i>	51.3	46.0	16.0	50.3	46.7	37.5
Llama-3.1-8B	min- <i>p</i>	51.2	50.8	39.5	51.2	50.2	43.5
Mistral-7B	top- <i>p</i>	30.2	29.5	25.3	29.2	30.2	32.5
Mistral-7B	min- <i>p</i>	33.8	30.7	31.0	33.2	33.3	28.7
Qwen2.5-7B	top- <i>p</i>	65.5	62.8	62.7	63.7	62.7	61.3
Qwen2.5-7B	min- <i>p</i>	61.3	63.5	59.7	64.2	63.0	62.3
Gemma-3-12B	top- <i>p</i>	65.7	66.7	63.8	65.8	65.0	65.2
Gemma-3-12B	min- <i>p</i>	65.3	67.0	65.5	65.3	63.7	63.8
Qwen3-1.7B	top- <i>p</i>	47.2	47.0	47.2	47.7	46.8	43.5
Qwen3-1.7B	min- <i>p</i>	47.0	47.7	46.2	45.7	46.2	49.0
Qwen3-4B	top- <i>p</i>	62.8	61.3	62.8	62.9	62.7	61.7
Qwen3-4B	min- <i>p</i>	60.9	62.7	62.9	61.3	63.6	61.5
Qwen3-8B	top- <i>p</i>	60.5	61.3	60.5	60.7	59.5	60.5
Qwen3-8B	min- <i>p</i>	61.0	59.2	59.7	61.3	60.7	59.3

Table 9: Mean distinct parsed answers per item ($K=3$, pooled over quantization).

Model	GSM8K		MMLU-Pro	
	$T=0.7$	$T=1.3$	$T=0.7$	$T=1.3$
Llama-3.1-8B	1.51	2.33	1.71	1.28
Mistral-7B	2.21	2.48	1.97	2.21
Qwen2.5-7B	1.10	1.27	1.44	1.57
Gemma-3-12B	1.12	1.16	1.33	1.45
Qwen3-1.7B	1.41	1.43	1.67	1.73
Qwen3-4B	1.05	1.14	1.33	1.38
Qwen3-8B	1.09	1.13	1.41	1.40

Table 10: Mechanism probes per model (Q8, 100 frozen prompts). Drop is the MMLU-Pro plain-temperature $T_{0.7} \rightarrow 1.3$ accuracy drop, pooled over quantization.

Model	H_{lb}	flat%	esc@1.3	Δ_{esc}	ΔH	drop (pp)
Llama-3.1-8B	0.329	7.1	3.26	+4.20	+0.110	+40.8
Mistral-7B	0.322	6.8	3.40	+1.19	+0.044	+8.0
Qwen2.5-7B	0.257	4.8	1.97	+1.00	+0.049	+4.8
Gemma-3-12B	0.180	3.0	1.44	+1.42	+0.055	+3.7
Qwen3-1.7B	0.150	2.1	0.36	-0.00	+0.012	+0.7
Qwen3-4B	0.125	1.5	0.22	+0.21	+0.021	+0.7
Qwen3-8B	0.216	3.8	0.71	+0.22	+0.024	+1.0